

Sales Analysis using Python in Excel

Dataset Description

The dataset contains the following columns:

- Date: Format - YYYY-MM-DD (converted to DD-MM-YYYY)
- Customer: Name of the customer
- City: Bengaluru, Mangaluru, Mysuru, Hubli
- Mode: Online or Cash
- Quantity: Number of items purchased
- Sales: Total bill amount

Analysis Steps

Sheet 1: 'Sales Data' - Raw sales data with 19,577 rows.

Sheet 2: 'Python'

Data Import & Initial Exploration:

```
df = xl("Sales Data!A1:F19577", headers=True)
df[["Quantity", "Sales"]].describe()
```

Dynamic Grouping with Dropdowns:

```
df.groupby(xl("D3")).Sales.sum()
```

Payment Mode Description:

```
df.Mode.describe()
```

Time-based Grouping (Monthly, Weekly, Daily):

```
df.groupby(pd.Grouper(key="Date", freq=xl("G3"))).Sales.sum()
```

Summary Statistics:

```
df[["Sales"]].sum()
df[["Quantity"]].sum()
```

AOV by City & Mode:

```
df.groupby("City").apply(lambda x: x.Sales.sum() / x.Quantity.sum())
df.groupby("Mode").apply(lambda x: x.Sales.sum() / x.Quantity.sum())
```

Data Visualization

Sheet 3: 'Data Visualization'

1. Line Plot: Sales Over Time (M/W/D)

```
Showchart = df.groupby(pd.Grouper(key="Date", freq=xl("B2"))).Sales.sum()
Showchart.plot(x="Date", y="Sales", kind="line")
```

2. Pie Chart: Based on Mode or City

```
Showchart = df.groupby(xl("B3")).Quantity.sum()
Showchart.plot(x=xl("B3"), y="Quantity", kind="pie")
```

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3. Bar Plot: Sales by Mode or City

```
sns.barplot(x=xl("B4"), y="Sales", data=df)
plt.show()
```

4. Bar Plot: AOV by City

```
aov_city = df.groupby("City").apply(lambda x: x.Sales.sum() / x.Quantity.sum()).reset_index(name='AOV')
aov_city.plot(x="City", y="AOV", kind="bar", title="Average Order Value by City", legend=False)
```

5. Bar Plot: AOV by Mode

```
aov_mode = df.groupby("Mode").apply(lambda x: x.Sales.sum() / x.Quantity.sum()).reset_index(name='AOV')
aov_mode.plot(x="Mode", y="AOV", kind="bar", title="Average Order Value by Mode", legend=False)
```

Tools Used

- Excel (Data Entry, Formatting, Python in Excel)
- Python (Pandas, Seaborn, Matplotlib)
- Visualization: Embedded plots within Excel

Key Insights

- Total and average sales can be explored by city and payment mode.
- AOV comparison shows differences in customer behavior by city and payment type.
- Monthly, weekly, and daily trends are visible using dynamic grouping.